User Manual

1. Overview This repository provides an intelligent configuration performance learning tool that utilizes Decision Tree and Random Forest algorithms to predict software system performance, automatically comparing them against a Linear Regression baseline.

2. Repository Structure

1. Random Forest.py: Evaluates the Random Forest model vs. Baseline (Linear Regression).
2. Decision Tree.py: Evaluates the Decision Tree model vs. Baseline.
3. Visualization.py: Reads the text log outputs and generates a summary bar chart.
4. datasets/: The root directory for your configuration data. It must contain subfolders for each system (e.g., xz, h2, dconvert, x264), which in turn contain .csv files.

3. Step-by-Step Usage Guide

Step 1: Data Preparation Ensure your configuration data is in .csv format. Place the CSV files inside their respective system folders under the datasets/ directory (e.g., datasets/xz/data.csv). The target performance metric must be the last column in the CSV file.

Step 2: Run the Model Evaluations Execute the evaluation scripts from your terminal or IDE (e.g., VS Code):

1. Run python "Decision Tree.py" to test the Decision Tree model.
2. Run python "Random Forest.py" to test the Random Forest model. The console will output the MAPE, MAE, RMSE, and the p-value for statistical significance.

Step 3: Generate Visualizations (Optional but Recommended) To visualize the overall performance across different systems:

1. Copy the entire console output from Decision Tree.py and paste it into a new text file named exactly Decision tree results.txt in the root directory.
2. Copy the entire console output from Random Forest.py and paste it into a new text file named exactly Random Forest results.txt in the root directory.
3. Run python Visualization.py. The script will automatically parse the text files, calculate the average errors, and generate a system_comparison_3bars.png chart in your current directory.